

Gut inflammation in chronic fatigue syndrome

Chronic fatigue syndrome (CFS) is a debilitating disease characterized by unexplained disabling fatigue and a combination of accompanying symptoms the pathology of which is incompletely understood. Many CFS patients complain of gut dysfunction. In fact, patients with CFS are more likely to report a previous diagnosis of irritable bowel syndrome (IBS), a common functional disorder of the gut, and experience IBS-related symptoms. Recently, evidence for interactions between the intestinal microbiota, mucosal barrier function, and the immune system have been shown to play a role in the disorder's pathogenesis. Studies examining the microecology of the gastrointestinal (GI) tract have identified specific microorganisms whose presence appears related to disease; in CFS, a role for altered intestinal microbiota in the pathogenesis of the disease has recently been suggested. Mucosal barrier dysfunction promoting bacterial translocation has also been observed. Finally, an altered mucosal immune system has been associated with the disease. In this article, we discuss the interplay between these factors in CFS and how they could play a significant role in GI dysfunction by modulating the activity of the enteric nervous system, the intrinsic innervation of the gut. If an altered intestinal microbiota, mucosal barrier dysfunction, and aberrant intestinal immunity contribute to the pathogenesis of CFS, therapeutic efforts to modify gut microbiota could be a means to modulate the development and/or progression of this disorder. For example, the administration of probiotics could alter the gut microbiota, improve mucosal barrier function, decrease pro-inflammatory cytokines, and have the potential to positively influence mood in patients where both emotional symptoms and inflammatory immune signals are elevated. Probiotics also have the potential to improve gut motility, which is dysfunctional in many CFS patients.